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New Early Neoproterozoic Paleomagnetic Constraints of the Northwestern China Blocks on the Periphery of Rodinia

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Key Points:

- A new paleomagnetic pole was obtained from eight early Neoproterozoic mafic dikes in the northern Altyn Tagh block
- The pole was dated at 895.3 ± 7.7 Ma using baddeleyite U-Pb SHRIMP methodology
- The adjacent south Tarim-Altyn Tagh-Qaidam-Qilian block and north China craton were located at NW side of Laurentia on the periphery of Rodinia

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Recent studies highlight the debates about the role of the northwestern China blocks, for example, north Tarim block, south Tarim block (STB), Altyn Tagh block (ATB) and Qaidam-Qilian block (QQB) during the assembly of Rodinia. We herein present a combined paleomagnetic, geochronological, and geochemical study on 16 mafic dikes in the northern ATB. After removing a recent viscous remanent magnetization, three characteristic remanent magnetization components (CHs) were isolated. CH1 was identified in eight NW-trending diabase dikes, one of which was newly dated at 895.3 ± 7.7 Ma through use of SHRIMP baddeleyite U-Pb methodology. It yields a mean direction (D/I) of 123.3°/−42.2° (α_{95} = 12.1°). The paleopole (40.9°S/184.6°E, A_{95} = 11.7°) was calculated by averaging eight virtual geomagnetic poles (VGPs) from each dike. A partial baked-contact test supports the primary origin of CH1. CH2 was isolated from three E-W trending diabase dikes, pointing northwest with considerably shallower downward inclinations. CH3 was obtained from five lamprophyre dikes, directing southeast with moderate to shallow upward inclinations. However, the VGPs of either CH2 or CH3 are insufficient to average a precise mean pole. Geochemical analyses also reveal petrogenetic discrepancy among the dikes with distinct CHs. Combining the new ~895 Ma pole of CH1 with the global paleomagnetic database and being aided by geological evidence, we propose that the STB-ATB-QQB and the north China craton were located along the northwestern side of Laurentia in Rodinia at ~895 Ma, and suggest the connection might have been intact until ~780 Ma.

Plain Language Summary The paleogeographic positions of the northwestern China blocks, including the north Tarim block, the south Tarim block, the Altyn Tagh block (ATB) and the Qaidam-Qilian block in the Rodinia supercontinent, remain enigmatic and highly controversial. Here we present new paleomagnetic geochronological and geochemical data from three groups of the mafic dikes in the northern ATB. Our new results, in combination with other coeval paleomagnetic poles and geological evidence, support a specific proximity among the south Tarim, Altyn Tagh, Qaidam-Qilian blocks and the north China craton along the northwestern side of Laurentia between ~895 and ~780 Ma.

1. Introduction

The Precambrian Qaidam-Qilian block (QQB), Altyn Tagh block (ATB) and Tarim block (TB) occupy a large area in northwestern China, surrounded by Phanerozoic orogenic belts or regional strike-slip faults (Figure 1a). Their paleogeographic affiliations with the north China craton (NCC) and their positions in the Rodinia supercontinent are intensively debated. One group of ideas places the TB, ATB, and QQB in between Australia and Laurentia in Rodinia (Lu et al., 2008; Wen et al., 2017, 2018; Xu et al., 2015), and Wen et al. (2018) recently suggested that a large transform fault along the tentative boundary across the central TB-ATB-QQB promoted the final assembly of Rodinia by ~900 Ma. However, this model is challenged by loosely constrained Neoproterozoic TB-ATB-QQB linkage (Song & Li, 2019), and a recent paleomagnetic study from the northern TB (Zhao et al., 2021) which unraveled a profound paleolatitudinal difference between the north Tarim block (NTB) and south Tarim block (STB) at ~900 Ma. Alternatively, other models include the proximities of the TB, ATB, and QQB with the NCC during the Rodinian era (Pang et al., 2021; Wu et al., 2021; Zuza & Yin, 2017), but are limited to geological considerations. In this study, we report the new paleomagnetic, geochronological and geochemical results obtained from 16 mafic dikes in the northern ATB (Figure 1a). The new data, along with other coeval poles from the NTB, STB, and NCC, and available geological evidence, permits a new contiguous paleogeographic configuration among the STB, ATB-QQB, and NCC at the northwestern side of Laurentia along the periphery of Rodinia between ~895 and ~780 Ma.

2. Geological Setting and Sampling

The ATB is situated between the TB and the QQB, NW China, bounded to the north and south by the Paleozoic Dunhuang orogen and the Paleo-Mesozoic Kunlun orogen, and to the west and east by the Cenozoic Qiemo-Mingfeng and Altyn Tagh faults (Figure 1a). Traditionally, the ATB was considered a part of the TB in Precambrian (e.g., Lu et al., 2008). However, stronger tectonic correlations between the ATB and the QQB are supported by a growing body of recent geological investigations. After the restoration of ~400 km displacement along the Altyn Tagh fault, the ATB together with the QQB, formed a coherent tectonic unit in Precambrian time (Song & Li, 2019; Zuza & Yin, 2017). Similar Neoproterozoic tectono-thermal activity is widely documented across the ATB and the QQB, including ~1000-900 Ma metamorphism, ~1030-850 Ma arc magmatism and ~850-620 Ma rift-related magmatism (Hao et al., 2020; Peng et al., 2019; Zuza & Yin, 2017). The Archean-Paleoproterozoic basement is well exposed in the northern ATB and the northern Qaidam block, known as Milan Complex and Delingha Complex, respectively (Figure 1a). Both underwent ~2000-1900 Ma metamorphism and were intruded by ~1850 Ma mafic dikes (Lu et al., 2008; Zhang et al., 2014), showing similar early Precambrian geological history among these regions. In addition, recent geochronological studies of the Precambrian sedimentary rocks, mainly exposed in the Central-Southern ATB and Qilian block, reveal consistent detrital zircon signature with ~1300-900 Ma maximum deposition ages, abundant ~1800-1300 Ma ages and minor Archean-Paleoproterozoic ages (Figure 1a; Hao, 2021; Li et al., 2020; Wu et al., 2017) suggesting there had been likely coherency between the Precambrian sedimentary basins.

Our sampling region is located in the northern part of the ATB, where Archean-Paleoproterozoic high-grade metamorphic basement rocks dominate, here referred to as North Altyn Tagh terrane (NAT) (Figure 1b; Zhang et al., 2014). Recent studies identified numerous mafic dikes in the NAT, which have different geological features and ages. One type is apparently altered, which has been dated at 1867 ± 8 Ma (Figure 1b; zircon, U-Pb laser ablation inductively coupled plasma mass spectrometer (LA-ICPMS) method; Zhang et al., 2014), while another type is quite fresh in outcrops and thin sections, and has been dated at 884.6 ± 4.2 Ma (Figures 1b and 2; zircon, U-Pb LA-ICPMS method; Zuo & Long, 2022). The fresh dikes can be further subdivided based on field observations and petrographic features. We have identified three groups in the region (Figure 1b). The first group includes many wider (>5 m in width) diabase dikes, showing NW trends (Figures 2a–2d), medium-grained, being composed of plagioclase (~50%), clinopyroxene (~45%) infilled by minor Fe-Ti oxides (Figures 2h–2i). The second group contains a few significantly narrower (<0.5 m in width) diabase dikes, showing E-W trends and ophitic texture (Figures 2e and 2j). The third group consists of lamprophyre dikes (Figures 2f–2g), varying in trends, being composed of hornblende (~45%), plagioclase (~40%), clinopyroxene (~10%) and Fe-Ti oxides (Figures 2k–2m).

In this study, 147 paleomagnetic core samples were drilled from eight NW-trending diabase dikes, and 25 samples from three E-W trending diabase dikes, 42 samples from five lamprophyre dikes (Figure 1b). Other 33 core samples from the gneissic host rocks were collected for baked-contact tests. All the core samples were oriented using both a magnetic compass and a sun compass. The orientation differences between the two methods are generally not more than 1°, indicating there is no significant local magnetic anomaly. All sampled dikes are vertical or sub-vertical and fresh in the outcrops. Twelve representative dikes were sampled for geochemical analyses, including four NW-trending diabase dikes (AT01, 02, 13 and 14), three E-W trending diabase dikes (AT03-05), and five lamprophyre dikes (AT07-09 and 11–12). Four block samples were further collected from dikes AT01, 02, 13, and 14 for geochronological analyses.

3. Laboratory Techniques

3.1. Geochronology

Sufficient baddeleyite grains were obtained from the samples of dike AT01 (of the first group) through use of conventional density and magnetic separation. They were mounted in an epoxy resin mount, together with the Phalaborwa baddeleyite standards (Heaman, 2009). The mount was polished to expose the inside of the baddeleyite grains. Transmitted and reflected light micrographs, and cathodoluminescence (CL) images were used to examine the internal structures of the grains, in order to determine the appropriate points for in situ measurements. U-Pb isotopic analyses of the baddeleyite were undertaken on a sensitive high-resolution ion microprobe (SHRIMP) instrument at the Beijing SHRIMP Centre, China Academy of Geological Sciences, following the

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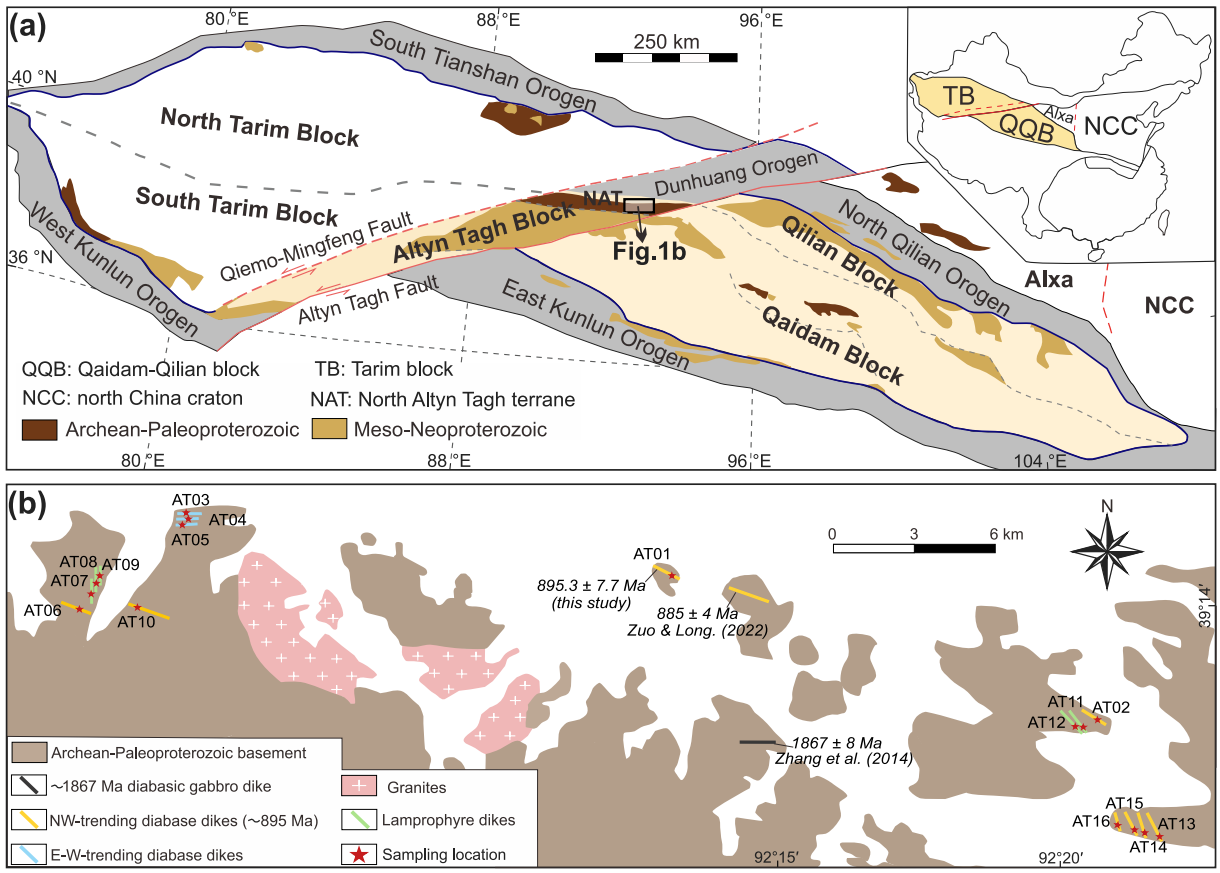


Figure 1. (a) Outline of the northwestern China blocks showing tectonic boundaries and the Precambrian outcrops (modified from Song & Li, 2019). (b) Simplified geological map of the North Altyn Tagh terrane (modified from Zuo & Long, 2022), showing the sampled NW-trending diabase dikes (golden lines), E-W trending diabase dikes (blue lines), and lamprophyre dikes (green lines).

method described in Williams (1998). The diameter of the analytical ion beam used is about 25 μm. Standard baddeleyite Phalaborwa was used to calibrate the Pb/U ratio and U concentration in the target baddeleyites. All data were processed using the Isoplot/Ex v. 4.15 program (Ludwig, 2012). Analytical errors for individual analysis in the data table (Table S1 in Supporting Information S1) are at 1σ level, whereas errors for weighted mean ages are quoted at 2σ level.

3.2. Paleomagnetism

The oriented paleomagnetic core samples were cut into 2.2-cm-long cylindrical specimens. In order to better understand the magnetic mineralogy and properties of remanence, rock magnetic experiments were conducted. Thermal demagnetization of triaxial isothermal remanent magnetization (IRM) (Lowrie test; Lowrie, 1990) was performed for representative specimens which were magnetized in turn along the Z, Y, and *X* axes in fields of 2.4, 0.4, and 0.12 T, and then underwent stepwise thermal demagnetization up to 690°C. Magnetic hysteresis loop parameters, including saturation magnetization (Ms), saturation remanence (Mrs), coercive force (Hc) and coercivity of remanence (Hcr), were measured for representative specimens through use of a Lakeshore 8600 Vibrating Sample Magnetometer installed at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction of Ministry of Natural Resources, CAGS, Beijing. All specimens were applied to stepwise thermal demagnetization in an ASC TD-48 furnace, the internal residual field in which is less than 10 nT. Nearly half of the parallel specimens were subjected to the alternating field (AF) demagnetization. Remanence was measured using an AGICO JR-6A spinner magnetometer for most specimens, and a 2G Enterprises 755-4K cryogenic magnetometer for the specimens with weak natural remanent magnetization intensities ($<1 \times 10^{-3}$ A/m). All the demagnetizations were carried out in a low residual magnetic field (<200 nT) shield room at the Paleomagnetism and Environmental Magnetism Laboratory in China University of Geosciences, Beijing. Magnetic components were determined

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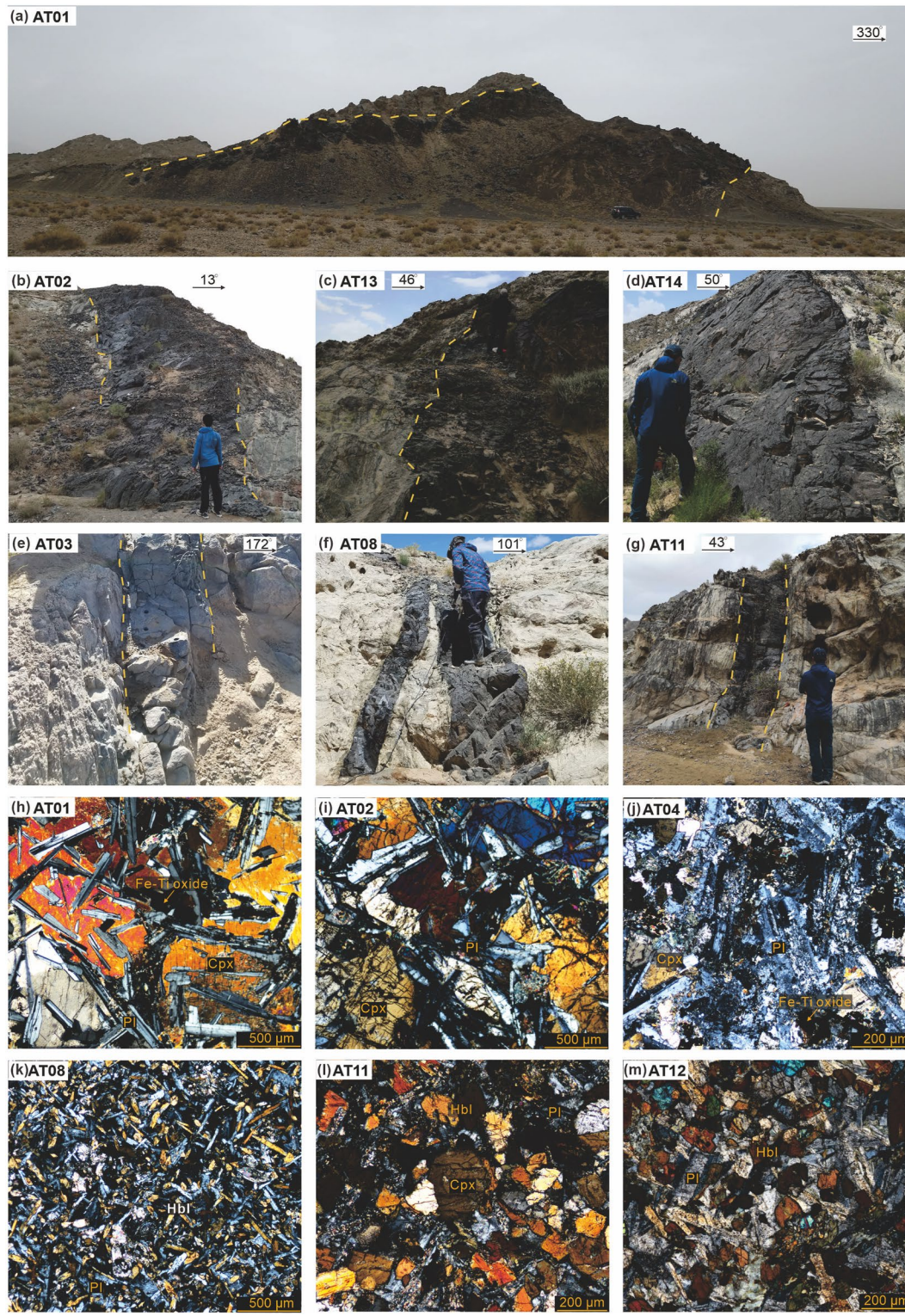


Figure 2. Representative outcrops and thin sections of sampled dikes. Photos of outcrops (a–d) NW-trending diabase dikes, (e) E-W trending diabase dikes, and (f–g) lamprophyre dikes. Yellow dashed lines represent the margins of dikes. Microphotographs of thin sections (h–i) medium-grained diabase (j) fine-grained diabase, and (k–m) fine-grained lamprophyre. Pl, plagioclase; Cpx, clinopyroxene. Hbl, hornblende.

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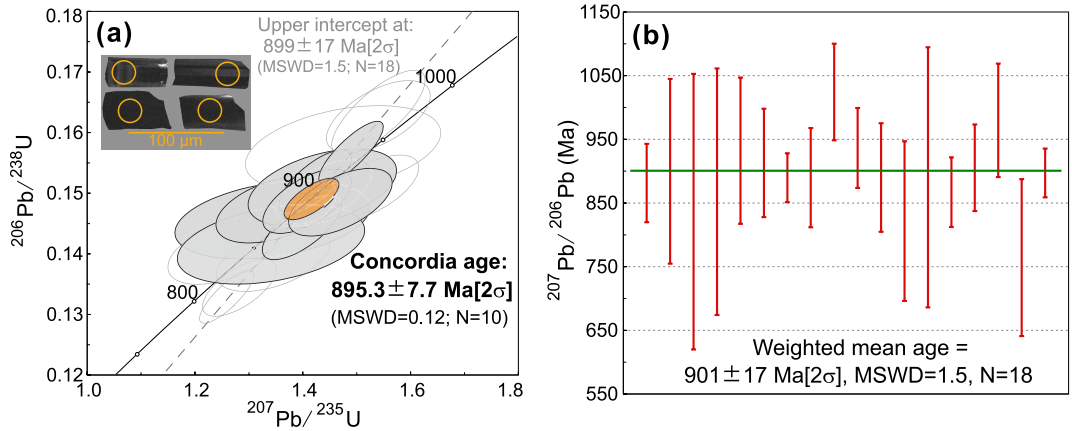


Figure 3. The baddeleyite U-Pb dating results of dike AT01. (a) Selected cathodoluminescence images and U-Pb concordia diagram of baddeleyite grains. (b) $^{207}Pb/^{206}Pb$ age plot of baddeleyite grains. The orange circles on baddeleyite grains represent the locations of sensitive high-resolution ion microprobe analyses (25 × 25 μm). The solid ellipses represent concordant data, and the hollow ellipses are discordant data. MSWD, mean square of the weighted deviations.

through use of the principal component analysis (Kirschvink, 1980). The mean directions were calculated using Fisher statistics (Fisher, 1953). The software packages of Enkin (1990) and Cogné (2003) were used to analyze the paleomagnetic data.

3.3. Geochemistry

The geochemical samples were crushed and pulverized to 200-mesh in an agate mortar. Major element oxides were measured by X-ray fluorescence spectrometry (XRF, PW4400) with relative standard deviations <0.5%. Powdered samples were completely dissolved in high-pressure Teflon bombs using a super-pure HF-HNO3 mixture for 48 hr at ~100°C. The trace-element concentrations of the samples were determined through use of an inductively coupled plasma-mass spectrometry (ICP-MS, PE300D). Duplicate analyses of samples and rock standards yielded relative standard deviations of <5% for most trace elements. All the whole-rock geochemical analyses were carried out at the National Research Center for Geoanalysis, Chinese Academy of Geological Sciences (CAGS), Beijing.

4. Results

4.1. Geochronological Results

The baddeleyite grains are tabular, with lengths of 50–80 μm and widths of 30–50 μm (Figure 3a). They show deep color or parallel striation, without a rim of polycrystalline metamorphic zircon in the CL images (Figure 3a). Eighteen spots, one spot on each baddeleyite grain, were analyzed. The 18 analyses yielded a weighted mean $^{207}Pb/^{206}Pb$ age at 901 ± 17 Ma (2σ; MSWD = 1.5) (Figure 3b), and an upper intercept age at 899 ± 17 Ma (2σ, MSWD = 1.5) (Figure 3a). Ten analyses therein are concordant and defined a concordia age at 895.3 ± 7.7 Ma (2σ, MSWD = 0.12) (Table S1 in Supporting Information S1; Figure 3a). The concordia age is more precise, not influenced by any "crystal orientation effect" (Wingate & Compston, 2000) or Pb loss. We thus consider it as the best estimate for the crystallization age of the dike AT01.

In a recent study, a weighted mean $^{206}Pb/^{238}U$ zircon age at 884.6 ± 4.2 Ma has been reported for a dike close to our sampled dike AT01 (Figure 1b; Zuo & Long, 2022). The age is identical to the baddeleyite age of dike AT01 within analytical error. The dike also has similar geochemical and structural features to that of the dike AT01, suggesting both the well-dated dikes are of the first group.

4.2. Rock Magnetic Results

The representative specimens from the NW-trending diabase dikes (of the first group) and the E-W trending diabase dikes (of the second group) show similar rock magnetic characteristics. Thermal demagnetization of the triaxial IRM (Lowrie, 1990) shows that the soft component (SC) (0.12 T) is dominant, with two-step

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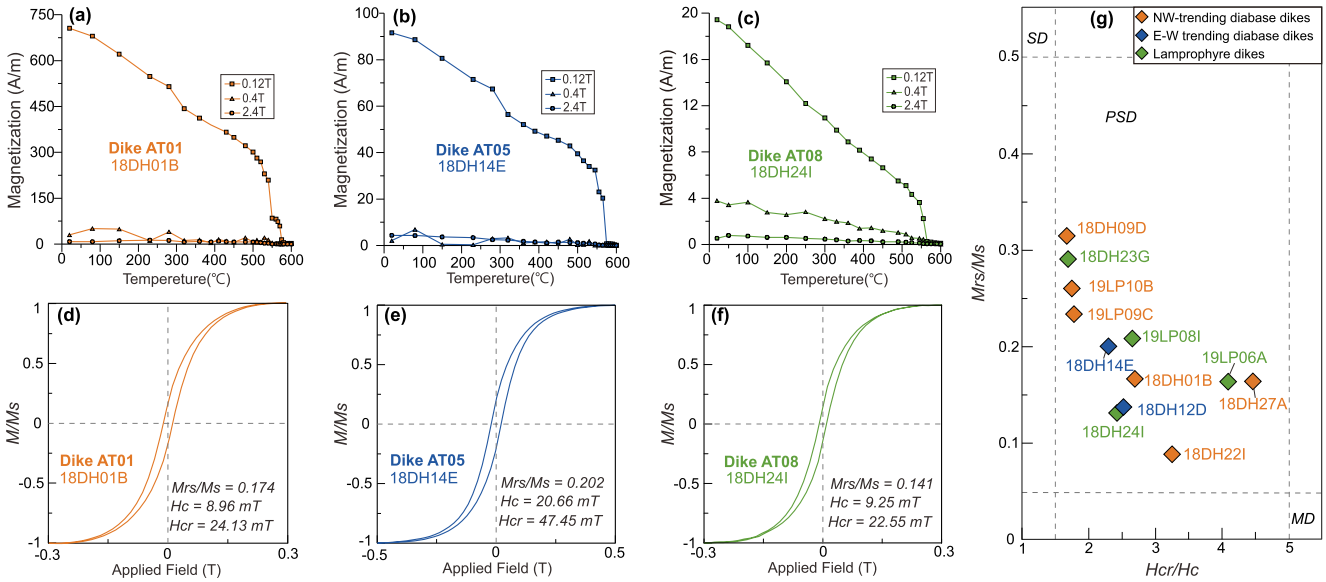


Figure 4. Rock magnetic results of representative samples. (a–c) Thermal demagnetization of the three-axis isothermal remanent magnetization. (d–f) Wasp-waisted magnetic hysteresis loops. (g) Day plot for samples from 12 dikes showing their PSD domain magnetite and poor-titanium titanomagnetite behaviors. Samples from the NW-trending diabase dikes, the E-W trending diabase dikes, and the lamprophyre dikes are marked as orange, blue, and green, respectively. Mrs, saturation remanence; Ms, saturation magnetization; Hcr, coercivity of remanence; Hc, coercivity; SD, single-domain; PSD, pseudo-single domain; MD, multidomain.

drops at ~550°C and ~580°C (Figures 4a–4b), indicating a major contribution of low-coercivity magnetite and poor-titanium titanomagnetite in the specimens. The hysteresis loops are wasp-waisted with saturation below 500 mT and low coercivity less than 30 mT (Figures 4d–4e), suggesting the dominance of low-coercivity magnetic carriers. On the Day plot, the hysteresis data for all analyzed specimens fall into the pseudo-single domain (PSD) region (Figure 4g; Day et al., 1977). These results collectively demonstrate that the paleomagnetically stable PSD magnetite and poor-titanium titanomagnetite are the major magnetic carriers of the diabase dikes.

The stepwise thermal demagnetizations of the triaxial IRM (Lowrie, 1990) of the lamprophyre dike specimens show the domination of SC (0.12 T), but with lower unblocking temperature at ~560°C (Figure 4c), indicating a domination of low-coercivity poor-titanium titanomagnetite. The wasp-waisted hysteresis loops show the specimens are saturated below 300 mT with coercivity less than 20 mT (Figure 4f). The ratios of Mrs/Ms and Hcr/Hc fall in the PSD region of the Day plot (Figure 4g; Day et al., 1977). These rock magnetic properties suggest that PSD poor-titanium titanomagnetite is the main magnetic carrier of the lamprophyre dike specimens.

4.3. Paleomagnetic Results

4.3.1. The NW-Tending Diabase Dikes (The First Group)

A total of 147 specimens from the eight NW-tending diabase dikes were subjected to the stepwise thermal demagnetization. Two paleomagnetic components were identified. A SC can be determined below 300°C in most cases (Figures 5a, 5c–5d, and Figures S2a–S2d in Supporting Information S1). The SC directions in situ are close to the local geocentric axial dipole field (Figure S3 in Supporting Information S1). It is hence interpreted as the viscous remanent magnetization (VRM) of the local recent geomagnetic field (RGF; $D = 0°$, I = 58.5°). A characteristic component (CH) was successfully isolated in steps between ~450°C and ~580°C (Figures 5a, 5c–5d, and Figures S2a–S2d in Supporting Information S1), which are consistent with the rock magnetic results. The CH directions of the eight dikes point southwest up with moderate inclinations, and yield a mean direction of $D = 123.3°$, I = −42.2° ($\alpha_{95} = 12.1°$, $n = 8$) (Table 1; Figure 6a). The CH can also be isolated between ~20 and ~70 mT, through use of AF demagnetization (Figure 5b, and Figures S2e–S2f, and Table S2 in Supporting Information S1), which strongly strengthens its stability. We name the CH of the eight NW-tending diabase dikes (of the first group) as CH1.

Host rocks of dikes AT01 and AT10, both yielding CH1, were sampled for baked-contact test. The baked gneissic samples were obtained within 0.5 m away from the contact, and the unbaked samples were collected at

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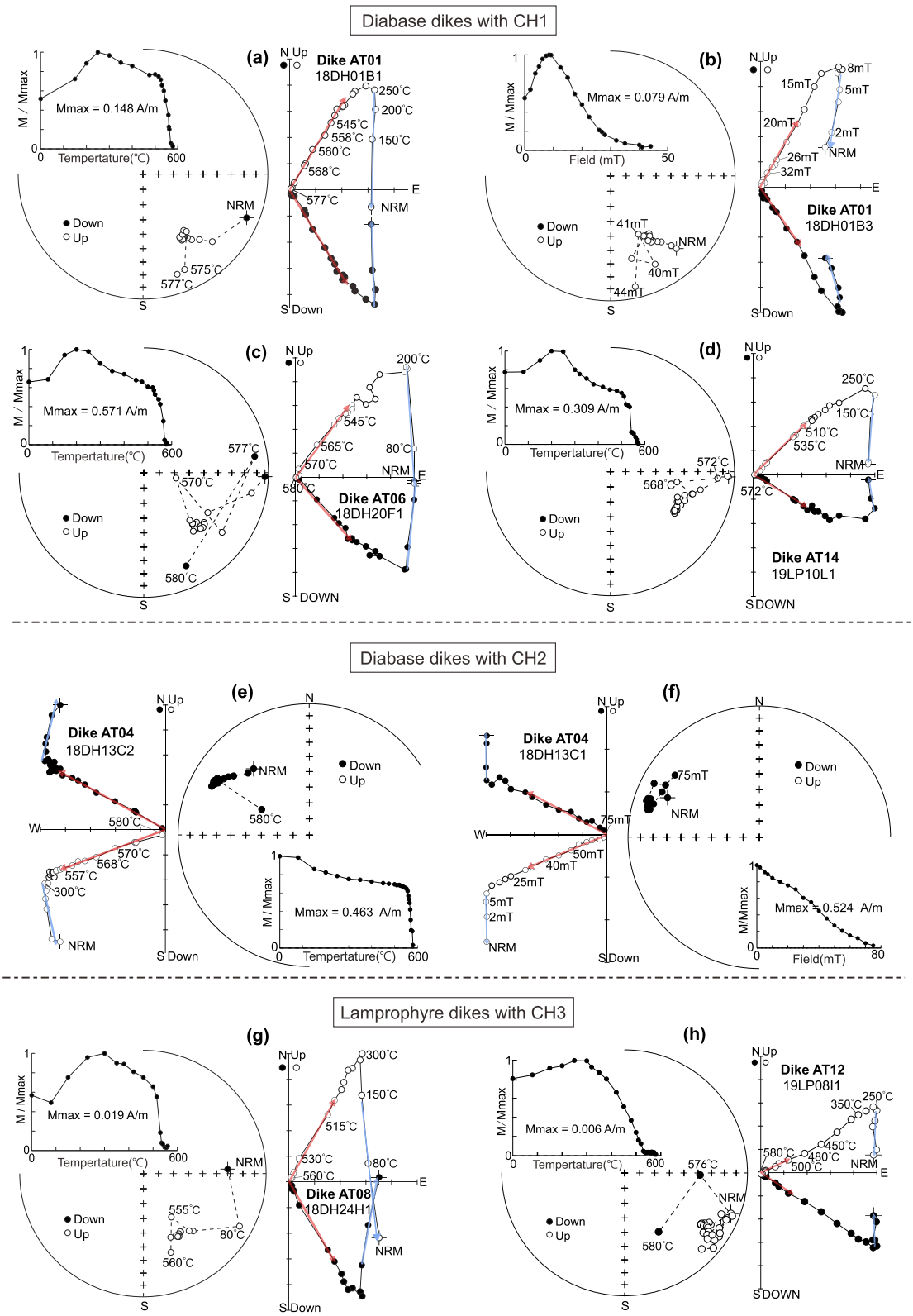


Figure 5. Orthogonal and equal-area projections (in geographic coordinates) and normalized magnetization intensity decay curves of representative samples from the dikes. (a–d) Diabase dikes with CH1. (e–f) Diabase dikes with CH2. (g–h) Lamprophyre dikes with CH3. Least-squares fits for the SC and CH are shown as blue and red arrows, respectively. M, magnetization; M_{max}, maximum magnetization; NRM, natural remanent magnetization.

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Table 1
Paleomagnetic Results of the Sixteen Mafic Dikes in the Northern Altyn Tagh Block

Dike	Lithology	Site	Width (m)	Trend/ Dip (°)	n/N	Slat (°N)	Slong (°E)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	Plong (°E)	d*p*/d*m* (°)
AT01	Diabase	18DH01-06	~120	295/70	43/54	39.2500	92.2184	145.1	−43.6	143.0	1.8	−57.7	167.5	1.4/2.2
AT02	Diabase	18DH09-11	~6	300/81	20/21	39.2019	92.3458	129.8	−23.7	112.5	3.1	−38.3	165.4	1.8/3.3
AT06	Diabase	18DH20-22	~15	290/82	21/22	39.2344	92.0478	126.9	−54.9	29.4	6.0	−48.2	194.1	6.0/8.5
AT10	Diabase	18DH27	~10	286/88	9/13	39.2342	92.0501	106.6	−67.2	47.2	7.6	−38.8	219.8	10.5/12.6
AT13	Diabase	19LP09	~7	335/81	4/7	39.1681	92.3569	131.2	−26.5	48.2	13.4	−40.4	165.9	7.9/14.5
AT14	Diabase	19LP10	~6	340/84	8/10	39.1673	92.3561	116.0	−47.0	102.2	5.5	−36.7	191.1	4.6/7.1
AT15	Diabase	19LP11	~15	339/80	7/10	39.1675	92.3554	112.1	−32.8	153.9	4.9	−28.1	183.1	3.1/5.5
AT16	Diabase	19LP12	~20	330/76	9/10	39.1683	92.3544	111.1	−37.7	67.2	6.3	−29.2	186.8	4.4/7.4
Mean direction of CH1					**8 (dikes)**			**123.3**	**−42.2**	**22.0**	**12.1**			
Mean pole of CH1					**8 (dikes)**							**−40.9**	**184.6**	**A_{95} = 11.7°**
														K = 23.2
AT03	Diabase	18DH12	~0.4	272/72	8/9	39.2617	92.0697	296.6	14.5	113.2	5.2	25.1	350.5	2.7/5.3
AT04	Diabase	18DH13	~0.5	265/78	6/8	39.2616	92.0697	295.8	19.6	117.7	6.2	26.3	353.4	3.4/6.5
AT05	Diabase	18DH14	~0.5	262/80	8/8	39.2616	92.0698	292.7	17.1	363.7	2.9	23.1	354.3	1.6/3.0
Mean for CH2					**3 (dikes)**			**295.0**	**17.1**	**632.0**	**4.9**			
AT07	Lamprophyre	18DH23	~0.3	359/73	6/6	39.2342	92.0481	147.3	−48.4	302.8	3.9	−61.4	172.0	3.4/5.1
AT08	Lamprophyre	18DH24	~0.4	344/81	6/6	39.2342	92.0482	150.4	−38.9	405.4	3.3	−59.5	156.4	2.3/3.9
AT09	Lamprophyre	18DH25	~0.3	359/75	8/10	39.2344	92.0483	150.9	−48.6	140.4	4.7	−64.3	168.9	4.1/6.2
AT11	Lamprophyre	19LP06	~1	320/83	9/10	39.1981	92.3425	134.0	−14.5	87.5	5.5	−37.9	157.1	2.9/5.6
AT12	Lamprophyre	19LP08	~1	315/85	8/10	39.1979	92.3425	121.2	−17.8	164.5	4.3	−29.8	169.0	2.3/4.5
Mean for CH3					**5 (dikes)**			**139.2**	**−34.3**	**17.0**	**19.1**			

Note. n/N, number of samples for statistics/total samples demagnetized; Slat/Slong, sample locality latitude/longitude; D/I, declination/inclination; k/K, fisher precision parameter for mean direction/pole; α_{95}/A_{95}, radius of circle of 95% confidence about the mean direction/pole; Plat/Plong, Latitude/Longitude of VGP and mean pole. d*p*/d*m*, semi-axes of elliptical error around the pole at a probability of 95%. CH, characteristic remanence magnetization. The bold values are part of the most important data of this paper.

a distance of ~50–100 m away from the contact (Figures 7a–7b). Most gneissic specimens yield a VRM of the RGF (Figure S4 in Supporting Information S1) as the SC of remanence. Three baked specimens of dike AT01 and four baked specimens of dike AT10 yield a CH, determined between 500°C and 580°C, being similar to CH1 (Figures 7d–7e and 7g–7h). Unfortunately, the CH directions of the unbaked specimens distribute randomly (Figures 7g and 7h). We considered that the results partially passed a baked-contact test, supporting the interpretation that the CH1 is a primary thermal remanent magnetization acquired during dike emplacement at ~895 Ma.

A mean paleomagnetic pole (40.9°S, 184.6°E, A_{95} = 11.7°) was calculated by averaging the eight virtual geomagnetic poles (VGPs) of each dike (Figure 6c; Table 1). Both the N-dependent A_{95} envelope and the scatter of the VGPs were calculated to evaluate the paleosecular variation (PSV). The A_{95} falls within an N-dependent A_{95} envelope (5.22°, 22.12°) at the 95% confidence interval (Deenen et al., 2011, 2014), and the S_B of the eight VGPs is 16.7° with 95% confidence interval of (13.4°, 20.0°), which overlaps the excepted range of (10.1°, 14.8°) based on the PSV model for 1.0–2.2 Ga period, while is slightly larger than the excepted range from 11.1° to 13.2° according to the PSV model for 0.5–1.5 Ga (reference point at 39.2°N, 92.2°E with a paleolatitude of 26.0°) (Smirnov et al., 2011; Veikkolainen & Pesonen, 2014). The results suggest that the pole is free of PSV influence. In addition, the average pole has no similarity to any younger paleomagnetic poles in the region. Overall, the new pole of CH1 satisfies six of the seven-point paleomagnetic data quality criteria (Meert et al., 2020) and can be considered as a high-quality pole of the ATB at ~895 Ma.

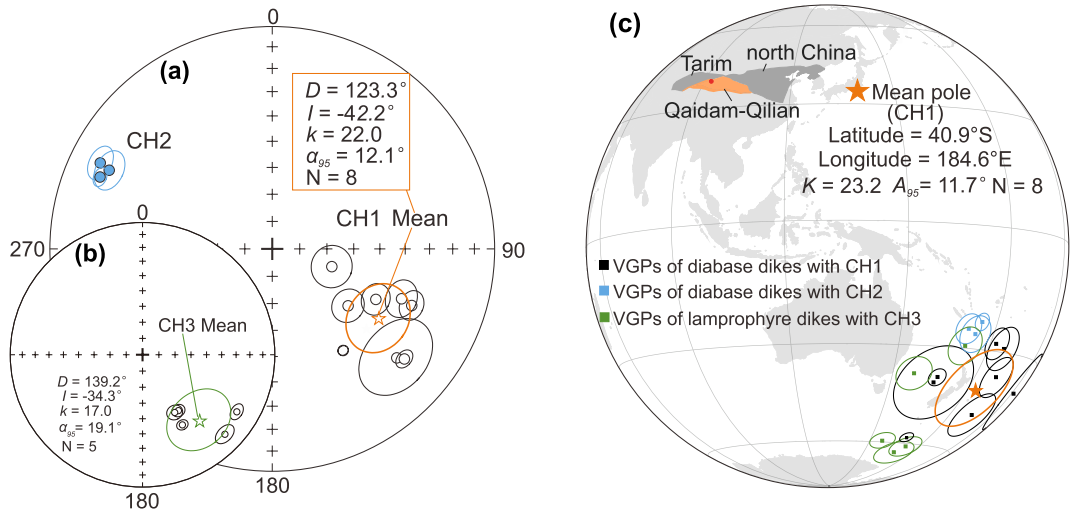


Figure 6. (a and b) Equal-area projection of the directions of CH1, CH2, and CH3. D, declination; I, inclination; k, Fisher precision parameter for mean direction; α_{95}, radius of 95% confidence cone of mean direction; *N*, number of dikes. (c) The positions of virtual geomagnetic poles (VGPs)s of diabase dikes with CH1 (black), diabase dikes with CH2 (blue), and lamprophyre dikes with CH3 (green), and the mean pole of CH1. The red dot in the Altyn Tagh-Qaidam-Qilian block shows the sampling location. K, Fisher precision parameter for pole; A_{95}, radius of 95% confidence cone of the pole; N, number of VGPs.

4.3.2. The E-W Tending Diabase Dikes (The Second Group)

A total of 25 specimens from the three E-W trending diabase dikes underwent stepwise thermal demagnetization. Their demagnetization behaviors are similar to those of the specimens from the NW-trending diabase dikes. An SC was defined below 300°C, and a CH was defined between ~450°C and ~580°C (Figure 5e and Figures S2g and S2h in Supporting Information S1). The SC also represents a recent VRM. The CH directions, however, point northwest down with significant shallower inclinations (Table 1; Figure 6a). The compatible CH could be isolated from the parallel specimens between ~30 and ~80 mT by AF demagnetization (Table S2 in Supporting Information S1; Figure 5f). Because it is obviously distinct from CH1, we name it CH2, representing the paleomagnetic signal of the E-W trending diabase dikes (of the second group) we sampled.

Three VGPs corresponding to each dike with CH2 were also calculated (Figure 6c; Table 1). However, they are insufficient to average out the PSV and thus will not be used further for the paleogeographic interpretation.

4.3.3. The Lamprophyre Dikes (The Third Group)

A total of 42 specimens from the five lamprophyre dikes were subjected to thermal demagnetization. An SC similar to the RGF (Figure S3 in Supporting Information S1) can be removed below 300°C. A CH was identified between ~450°C and ~560°C (Figures 5g–5h and Figures S2j and S2l in Supporting Information S1). The CH could be also isolated by the AF demagnetization between ~40 and ~120 mT (Table S2, Figures S2i, and S2k in Supporting Information S1). The unblocking temperature and coercivities of the CH are in line with the rock magnetic analyses (in Section 4.2). The CH directs southeast with shallow to moderate upward inclinations (Table 1; Figure 6b), which differs from CH1 or CH2 of the diabase dikes. We name it CH3, representing the paleomagnetic signals of the lamprophyre dikes. Due to the small number of the dikes studied, only five VGPs of each dike were calculated (Table 1, Figure 6c). They cannot average out the PSV too.

We also collected the gneissic host rock samples of the dike AT11 (~1-m-wide) within 0.5 m and at >8 m away from the dike's margin (Figure 7c) for baked contact test. The host rock specimens record an SC likely being the VRM of RGF (Figure S4 in Supporting Information S1), but yield scattered CH directions (Figure 7i). The results suggest that the host rocks were not affected by the narrower lamprophyre dike.

4.4. Geochemical Results

The whole-rock major and trace elements compositions of four CH1-bearing diabase dikes, three CH2-bearing diabase dikes and five CH3-bearing lamprophyre dikes are summarized in Table S3 in Supporting Information S1.

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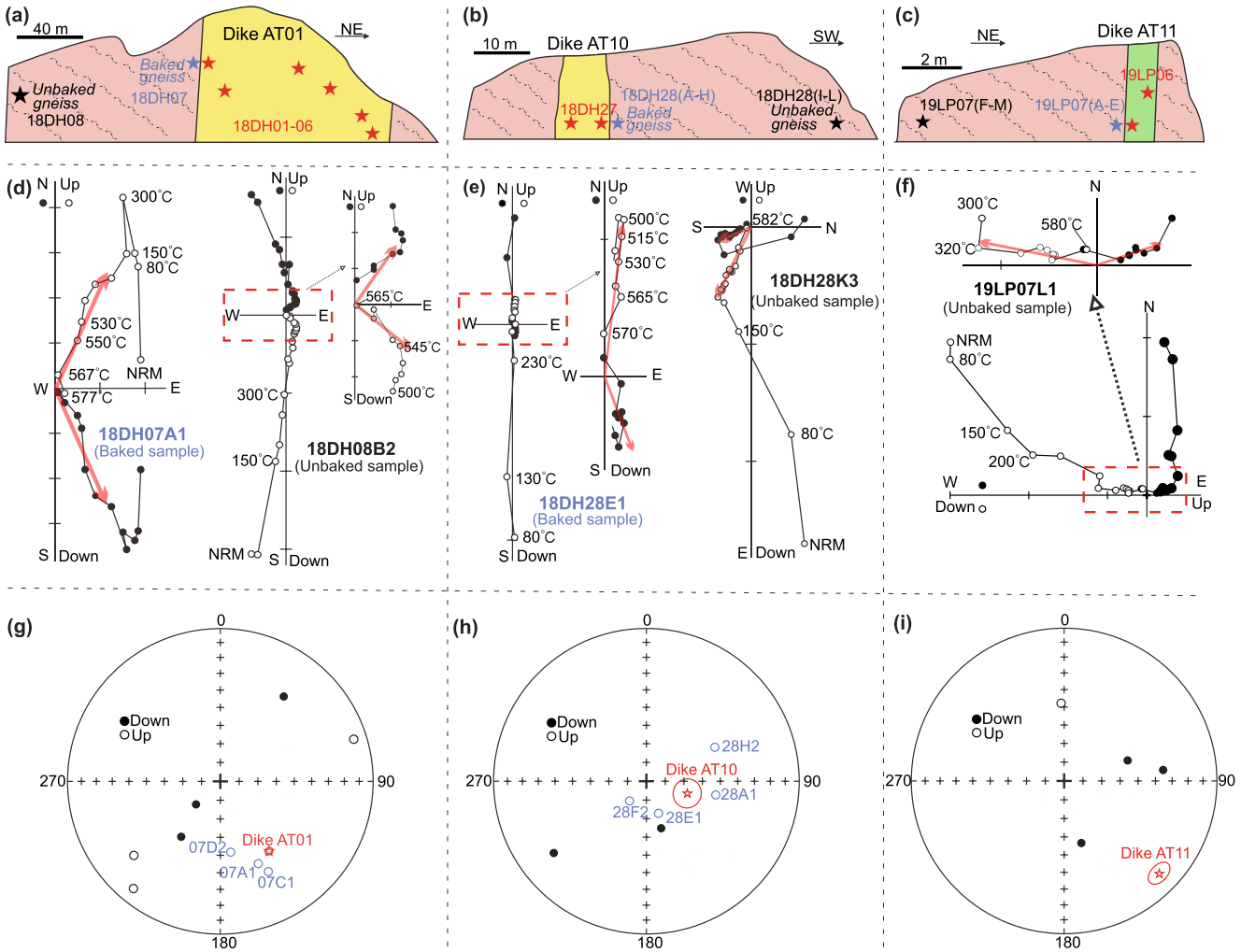


Figure 7. Baked-contact tests for two CH1-bearing diabase dikes (AT01 and AT10) and one CH3-bearing lamprophyre dike (AT11). (a–c) Geological sketches of outcrops showing the sampling locations (stars) of the three dikes, baked and unbaked host rocks. (d–f) Orthogonal projections of the representative baked and unbaked host rocks of the three dikes. The least-squares fit for the CH is shown as a red arrow. NRM, natural remanent magnetization. (g–i) Equal-area projections of the CH directions of the baked (blue) and unbaked host rocks (black), and mean CH directions of dikes (red). All projections are shown in geographic coordinates.

The major and trace elements concentrations show no obvious correlation with the loss on ignition (Figure S1 in Supporting Information S1), suggesting the chemical compositions of analyzed samples are not affected by weathering or alteration.

The CH1-bearing diabase dikes have a narrow range of SiO_2 (46.48–48.56 wt.%), variable $Fe_2O_3^T$ (10.51–13.34 wt.%), TiO_2 (1.24–3.21 wt.%) and MgO (6.24–8.12 wt.%) with Mg# values from 48.14 to 57.82. In the chondrite-normalized rare earth element plots, the samples display slightly enriched light rare earth element (LREE; chondrite-normalized La/Yb = 2.61–3.07) and weakly depleted heavy rare earth element (HREE; chondrite-normalized Gd/Yb = 1.66–1.85), which are similar to those of the ocean island basalts (OIB) to the enriched-mid ocean ridge basalts (E-MORB) (Figure 8a). In the primitive mantle-normalized multi-element spider plots, they show enrichment in Ba, U, Ta, and Zr concentrations, without pronounced negative Nb, Ta, and Ti anomalies (Figure 8b). The Th/Nb and TiO_2/Yb ratios range from 0.087 to 0.15, and 0.66 to 0.76, falling into the MORB-ocean plateau basalt-OIB array on the Th/Yb versus Nb/Yb and TiO_2/Yb versus Th/Nb plots, suggestive of an asthenospheric mantle source without apparent crustal input (Figure 9; Pearce et al., 2021). Our geochemical analyses of the CH1-bearing diabase dikes are consistent with the dated 884.6 ± 4.2 Ma diabase dike newly published in the same region (Zuo & Long, 2022), but show distinct geochemical features to the CH2-bearing diabase dikes and the CH3-bearing lamprophyre dikes.

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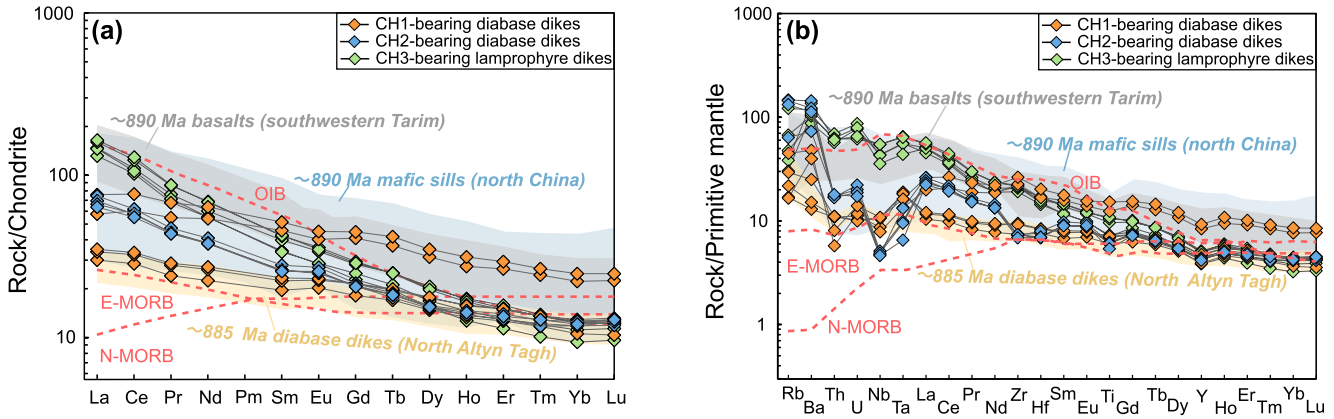


Figure 8. (a) Chondrite-normalized REE patterns. (b) Primitive mantle-normalized trace-element spider diagrams. Normalization data of ocean island basalts, E-MORB and N-MORB refer to Sun and McDonough (1989). The data of the ~885 Ma diabase dikes in the North Altyn Tagh are from Zuo & Long et al. (2022); data of the ~890 Ma basalts in the southwestern Tarim are from Zhang, Li, et al. (2019); data of the ~890 Ma mafic sills in the north China craton are from Peng et al. (2011) and references therein, and Zhang, Zhao, et al. (2016).

The CH2-bearing diabase dikes have slightly higher SiO_2 (48.54–48.92 wt.%) and $Fe_2O_3^T$ (13.38–13.46 wt.%), but lower TiO_2 (1.22–1.30 wt.%) and MgO (6.24–6.64 wt.%) with Mg# values (47.99–49.41). Their trace element compositions are featured by moderate enrichment in LREE (Chondrite-normalized La/Yb = 5.27–5.94) (Figure 8a) and significant negative high field strength elements (e.g., Th, Nb, Ta, Zr, and Ti) anomalies (Figure 8b). The Th/Nb and TiO_2/Yb ratios vary from 0.39 to 0.46, and 0.59 to 0.64, plotting into the subduction-modified lithospheric mantle (SZLM) array on the Th/Yb versus Nb/Yb and TiO_2/Yb versus Th/Nb diagrams, indicative of an SZLM-dominated source (Figure 9; Pearce et al., 2021).

The CH3-bearing lamprophyre dikes, however, are characterized by the lowest concentrations of SiO_2 (45.26–47.39 wt.%) and $Fe_2O_3^T$ (8.91–9.75 wt.%), but the highest concentrations of MgO (8.79–10.31 wt.%) and Mg# values between 65.05 and 69.62. Their chondrite-normalized REE patterns show highly enriched LREE (Chondrite-normalized La/Yb = 12.15–14.05) and depleted HREE (Chondrite-normalized Gd/Yb = 2.23–2.43), which are similar to those of OIB (Figure 8a). Their multi-element spider diagrams also exhibit enrichment in Ba, U, Ta, and Zr concentrations (Figure 8b). In the Th/Yb versus Nb/Yb and TiO_2/Yb versus Th/Nb diagrams, the samples plot within the enriched mantle-rich OIB part of the plume array (Figure 9), indicating the lamprophyre dikes were derived from the asthenospheric mantle with a high proportion of deep-recycled crustal materials (Pearce et al., 2021).

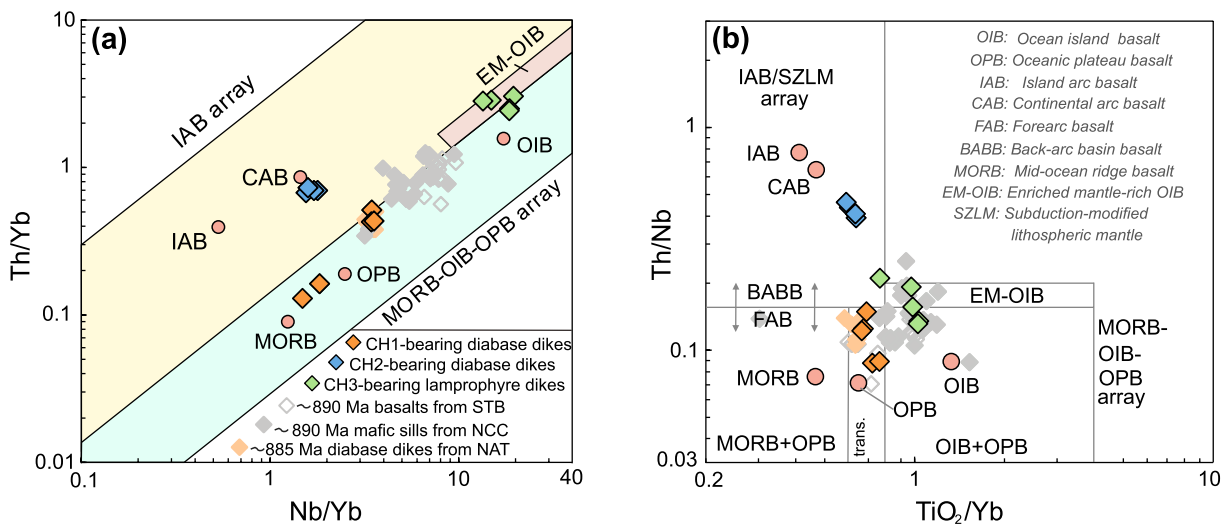


Figure 9. Major and trace elements discrimination diagrams. (a) Nb/Yb versus Th/Yb plot (Pearce et al., 2021). (b) TiO_2/Yb versus Th/Nb plot (Pearce et al., 2021). The sources of data are same as those in Figure 8.

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5. Discussion

5.1. Three Groups of the Sampled Dikes in the Northern ATB

Our geological investigations, paleomagnetic and geochemical analyses consistently indicate that the sampled 16 dikes can be divided into three groups. Group one, which contains the eight NW-trending diabase dikes with CH1, is characterized by consistent OIB-E-MORB-like REE and multi-element spider patterns with an asthenospheric mantle source, and is dated at 895.3 ± 7.7 Ma (baddeleyite SHRIMP U-Pb method, this study). Group two, which comprises the three E-W trending diabase dikes with CH2, is characterized by obvious negative Nb-Ta-Ti anomalies and an SZLM-dominated source. Group three is composed of the five lamprophyre dikes with CH3, showing OIB-like features and an asthenospheric mantle source with a large contribution of deep-recycled crustal materials.

In large scale comparison, the group one dikes are geochemically similar to the ~890 Ma Chulan-Dalian-Sariwon-Xuhuai mafic sills in the NCC (Peng et al., 2011 and references therein; Zhang, Zhao, et al., 2016), and the ~890 Ma Sailajiazitage basalts in the STB (Figures 8 and 9; Zhang, Li, et al., 2019). If there was then paleogeographic proximity among the ATB, NCC and STB, the group one dikes could be correlated with the ~890 Ma large igneous province (LIP) in the NCC and STB. However, the second-group diabase dikes and the lamprophyre dikes need further direct age constraints.

5.2. The Position of the Northwestern China Blocks in the Periphery of Rodinia

The new ~895 Ma pole places the sampling site (39.2°N, 92.2°E) at a paleolatitude of 26.0° ± 11.7°, confining the ATB to low latitudes, which is comparable to the coeval paleolatitudes of the NCC indicated by the paleomagnetic pole of the ~890 Ma mafic sills in the eastern NCC (Fu et al., 2015; Zhao et al., 2020) and that of the STB constrained by the ~890 Ma pole of the Sailajiazitage volcanics (Wen et al., 2018). Through matching these three high-quality poles, and adopting the recent paleogeographic model of NCC-NW Laurentia connection in Rodinia (Ding et al., 2020, 2021; Fu et al., 2015; Zhao et al., 2020, 2022), we herein propose a paleomagnetic best-fit location for the ATB that was in between the NCC and STB: the present eastern margin of the ATB facing the NCC's present western margin, and the present western side of the ATB being close to the STB (Figure 10a). Both the ATB and STB were attached on the NW Laurentia side of the Rodinia supercontinent together with the NCC, with the present northern margin of the STB facing an open ocean. Considering the tight tectonic correlations among the ATB and QQB, we suppose the entire ATB-QQB was sandwiched between the NCC and STB (Figure 10a).

Although paleogeographic interpretation of the new paleomagnetic data has the hemispherical ambiguity and the paleolongitudinal uncertainty, our model that emphasizes the proximity of the STB, ATB-QQB, and NCC, can be supported by the following geological correlations: (a) Contiguous early Paleozoic-Mesozoic geological records across the TB, ATB-QQB, and NCC after the restoration of Cenozoic deformation, suggesting these blocks were linked by at least the early Paleozoic (Zuza & Yin, 2017). (b) Highly similar upper Neoproterozoic lithostratigraphic sequence of the Qaidam block and NCC, which is marked by the post-Gaskiers glaciation, and the overlying siliciclastic depositions with the Ediacara-type fossil *Shaanxilithes* (Pang et al., 2021 and references therein). (c) Shared synchronous Neoproterozoic LIP fragments (e.g., the ~890 Ma LIPs across the ATB-QQB, NCC, and the STB, Peng et al., 2011; Zhang, Zhao, et al., 2016; Zhang, Li, et al., 2019; Zuo & Long, 2022). Both the ATB-QQB and NCC recorded two other phases of LIPs, including ~850-820 Ma LIP with consistent OIB-E-MORB type compositions formed in intraplate settings (e.g., the 846-821 Ma Yingfeng mafic dikes in the northern Qaidam block, Lu et al., 2008; Xu et al., 2016; the ~825 Ma Jinchuan mafic-ultramafic rocks and the ~830 Ma Luanchuan gabbro in the NCC, Li et al., 2005; Wang et al., 2011), and ~780 Ma LIP sourced from enriched mantle with contamination of continental crust (e.g., the ~780 Ma meta-mafic volcanics in the northern Qaidam block, Peng et al., 2019; the ~775 Ma Lulong mafic dikes in the NCC, Ding et al., 2021). (d) Comparable late Paleoproterozoic-Mesoproterozoic cover sequence and Paleoproterozoic magmatic-metamorphic events: the late Mesoproterozoic Tuolai Group in the Qilian block and the Suoerkuli Group in the ATB could be equivalent to the Dunzigou Group in the southwestern NCC and the Baiyinbaolage Formation in the northern NCC, based on similar detrital zircon age populations (Wu et al., 2021). The recently revised <1730-1640 Ma rift to passive margin strata in the northern Qaidam block—Quanji Group can be correlative with the Changcheng Group in the NCC (Wang et al., 2022; Zhang, Wang, et al., 2016), and the ~1500 Ma Sangzhutage Group in the southwestern

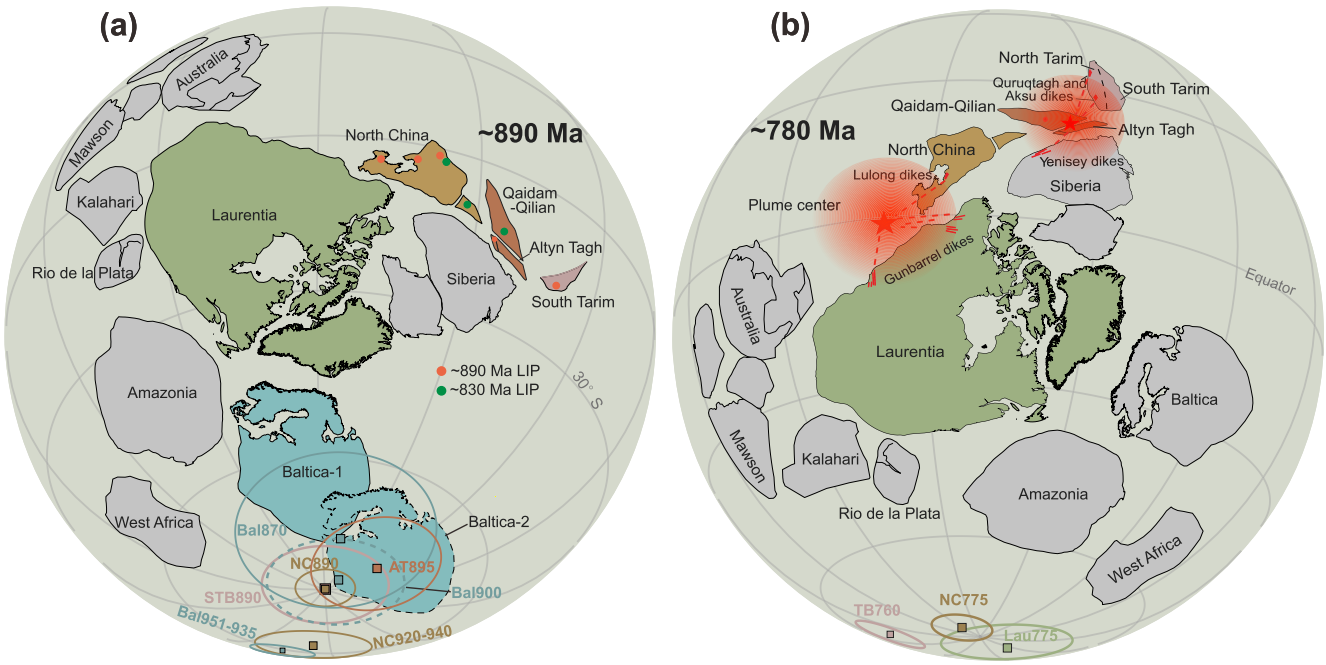


Figure 10. Paleogeographic reconstruction of Rodinia. (a) ~890 Ma reconstruction. The partnership among the south Tarim block (STB), Altyn Tagh block (ATB) and north China craton (NCC) is based on the fit of three coeval poles from each block. Blocks and their poles (listed in Table 2) are matched in color. The NCC-NW Laurentia connection is based on Ding et al. (2021). The baltica-1 position is adopted from the "right-way-up" model (Cawood & Pisarevsky, 2006). The baltica-2 position is based on the fit of new 900 Ma pole ("Bal900"; Kulakov et al., 2022) with the pole "NC890." The "tight fit" between Siberia and Laurentia is adopted from Evans et al. (2016). (b) ~780 Ma reconstruction. The relationship between the STB-ATB-QQB, NCC, and Laurentia at ~780 Ma is same as that of (a) (based on the poles "NC775," "Lau775," and "TB760" in Table 2), in the case of the Tarim block (TB) was united at this time. The plume centers are inferred from fitting ~780 Ma mafic dikes/volcanics around the TB-QQB-Siberia, and NCC-Laurentia, respectively. Parameters of the Euler rotation are listed in Table S4 in Supporting Information S1.

STB shares the same passive margin rock assemblage with that of the NCC as well (Zhang, Ye, et al., 2019). Moreover, both ~1900 Ma metamorphism and ~1800 Ma extensional magmatism had been widely reported from the southwestern STB, ATB-QQB, and NCC (Lu et al., 2008; Zhang et al., 2014; Zhang, Ye, et al., 2019; Zhang et al., 2021 and references therein).

Our model places the ATB-QQB next to the present western margin of Siberia craton (Figure 10a), if we adopt the "tight fit" model of the Laurentia-Siberia connection in Rodinia (e.g., Ernst et al., 2016; Evans et al., 2016). It does make sense from several geological observations available. Similar middle Neoproterozoic intraplate magmatism had been identified from both continents, that is, the ~790 Ma mafic dikes in the Yenisey Ridge (Likhanov & Santosh, 2017) and the ~780 Ma meta-mafic volcanic rocks in the northern Qaidam block (Peng et al., 2019). The recently revealed Grenville-aged (~1100-900 Ma) granitoid and metamorphic rocks in both the Yenisey Ridge (Likhanov et al., 2014) and the ATB-QQB (Peng et al., 2019; Song et al., 2012) may represent the late Mesoproterozoic to early Neoproterozoic convergence between western Siberia and the ATB-QQB.

Our reconstruction indicates that the western Laurentia along with the NCC, ATB-QQB, and STB were located in low latitudes at ~890 Ma (Figure 10a). It is also paleomagnetically permitted to place the STB on the outboard of western Laurentia; however, whether it was connected with Australia craton (Wen et al., 2018) is unclear because there is no coeval paleomagnetic pole from Australia. Recent paleomagnetic results from the Qigelekekuotan volcanics in the northern TB revealed considerably higher paleolatitudes of the NTB at ~900 Ma (Zhao et al., 2021), excluding its neighborhood relationship with western Laurentia. In conclusion, it is unlikely that a united TB played a linkage role between Australia and Laurentia (Wen et al., 2017, 2018) at ~900 Ma.

The reconstruction model can also be valid at ~780 Ma (Figure 10b), when the STB and NTB were likely united as one tectonic unit—TB (Zhang, Ye, et al., 2019; Zhang et al., 2021; Zhao et al., 2021). Three ~770 Ma paleomagnetic poles (TB760, NC775, and Lau775; Table 2) can support the same paleogeographic relationship between the STB and NCC as that in our ~890 Ma model. Both the middle Neoproterozoic plume-related

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Table 2
Selected Neoproterozoic Paleomagnetic Poles From the Northern Altyn Tagh Block, Tarim Block, North China Craton, Laurentia and Baltica

Pole name	Rock unit	Nominal age (Ma)	Plat (°N)	Plong (°E)	A_{95} (°)	1234567 (R)	Paleomagnetic references	Age references
Altyn Tagh block								
AT895	Northern Altyn Tagh ~895 Ma dikes	895.3 ± 7.7, UB	−40.9	184.6	11.7	1111101 (6)	This study	This study
Tarim block								
STB890	Sailajiazitage volcanics	~890, UZ	23.5	217.0	11.3	1111101 (6)	Wen et al. (2018)	Wang et al. (2015); Zhang, Li, et al. (2019)
TB760	Aksu ~760 Ma dikes	759 ± 7, UZ	19.0	128.0	6.5	1110111 (6)	Chen et al. (2004)	Zhang et al. (2009)
North China craton								
NC920-940	~920–940 Ma sills	920-940, UZ	−28.2	141.9	10.4	1111110 (6)	Fu et al. (2015) Zhao et al. (2020)	Zhao et al. (2020)
NC890	Huaibei 890 Ma sills	890 ± 14, UZ	−52.6	150.0	5.3	1111111 (7)	Fu et al. (2015)	Wang et al. (2012)
NC775	Lulong dikes	775 ± 5, UZ	−29.0	64.7	5.4	1110111 (6)	Ding et al. (2021)	Wang et al. (2016)
Laurentia								
Lau775	Gunbarrel LIP	774.93 ± 0.54, UZ	9.1	138.2	11.7	1110111 (6)	Recalculated by Eyster et al. (2020)	Milton et al. (2017)
Baltica								
Bal951-935	Blekinge-Dalarna dikes	951-935, UB, Ar	−2.6	239.6	5.8	1111111 (7)	Gong et al. (2018)	Gong et al. (2018)
Bal900	Western Telemark. Mafic xenoliths	~900, Ar	−46.2	209.4	12.0	1111001 (5)	Kulakov et al. (2022)	Kulakov et al. (2022)
Bal870	Rogaland Igneous Complex	869 ± 14, Ar	−45.9	238.4	18.1	1011001 (4)	Walderhaug et al. (2007)	Walderhaug et al. (2007)

Note. Dating method of age: Ar, ^{40}Ar-^{39}Ar; UB, U-Pb baddeleyite; UZ, U-Pb zircon; Pole name, paleopoles abbreviation used in the text and reconstructions; Plat/Plong, latitude/longitude of the pole; A_{95}, the radius of the circle of 95% confidence about the pole; the seven quality criteria and "R" factor was described by Meert et al. (2020).

magmatism in the northern China blocks, including the ~800-760 Ma Quruqtagh and Aksu dikes in the TB (Zhang et al., 2009), the ~780 Ma meta-mafic volcanics in northern Qaidam block (Peng et al., 2019), and the ~775 Ma Lulong dikes in the NCC (Wang et al., 2016) sourced from the SZLM, and the ~790 Ma Yenisey dikes along western Siberia with intraplate basalt geochemistry (Likhanov & Santosh, 2017), are comparable with that of the ~775 Ma Gunbarrel dikes along western Laurentia (Mackinder et al., 2019 and references therein). The geometry of these dike swarms could constitute the contemporary LIPs in western Laurentia and the NCC-ATB-QQB-TB (Figure 10b).

Some researchers also speculated on a long-lived proximity of the northwestern China blocks and NCC with Baltica (Zuza & Yin, 2017), based on the geological similarities among these blocks. The STB, ATB-QQB, and NCC, however, were located at low paleolatitudes at ~890 Ma, while the ~950-870 Ma paleomagnetic poles (Bal951-935, Bal900, Bal870; Table 2) place Baltica in significantly higher paleolatitudes. The current paleomagnetic data do not support a model that Baltica was close to the STB, ATB-QQB, and NCC, but they can be placed on the opposite sides of Laurentia instead (Figure 10a; Fu et al., 2015).

6. Conclusions

A new paleomagnetic pole was obtained from eight NW-trending diabase dikes in the northern ATB. One of the dikes was newly dated at ~895 Ma through use of baddeleyite U-Pb SHRIMP methodology. The new data together with global paleomagnetic database and aided by geological evidence, suggest that the neighboring STB, ATB-QQB and NCC were located along the NW side of Laurentia in Rodinia at ~895 Ma. The connection might be valid until ~780 Ma.

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Data of this paper are presented in Supporting Information S1, and also available for download through the Figshare (https://doi.org/10.6084/m9.figshare.21729173).

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